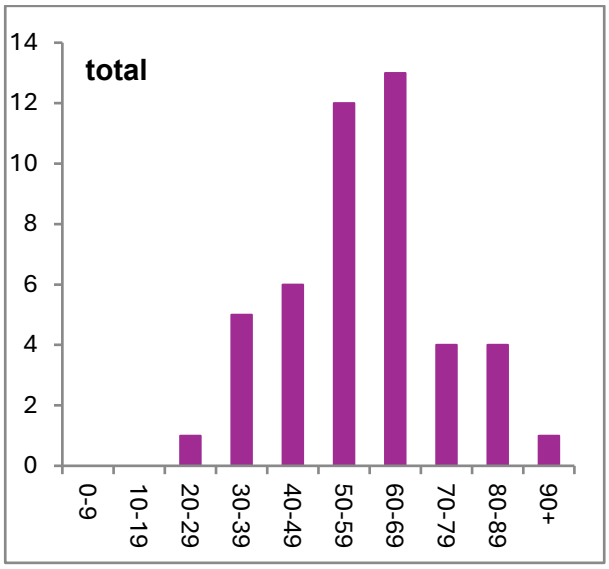
EART22201 Igneous Minerals and Processes

Practical test 15 December 2025 – class-level feedback

Overall: Thank you to the whole class for your hard work and high engagement over the semester. This really paid off in the practical test. Many answers demonstrated that you had paid careful attention in the lectures and practicals by correctly interpreting the questions and avoiding some common mistakes made by previous yeargroups.

Here is the overall marks distribution. The mean mark was 59%, and the standard deviation was 15%.

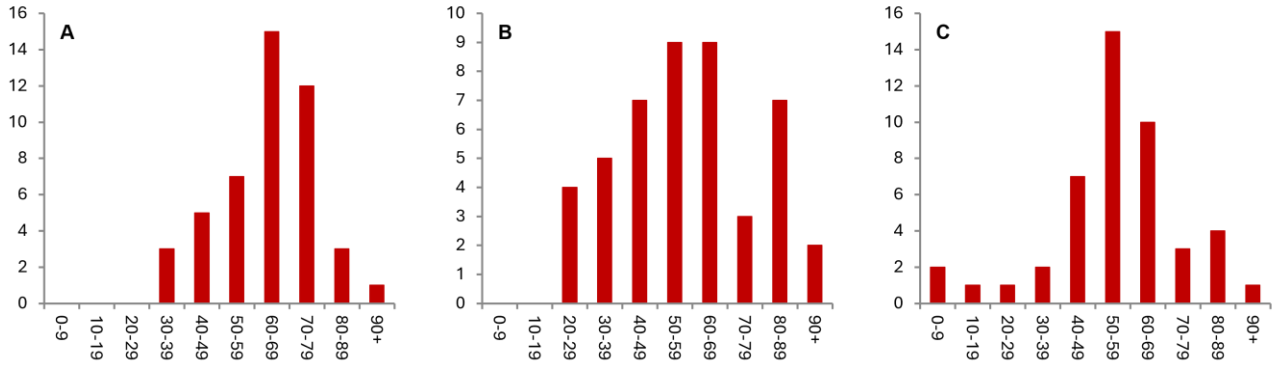


Section A asked you to examine two thin sections and answer some short questions about each one. The best answers provided neat sketches carefully labelled with the observations used to identify the minerals, and labels explaining key features of the rock texture. Common mistakes included forgetting to include a scale bar on sketches; misidentifying minerals but without providing any observations (so I could not award partial credit); and running out of time leaving some questions blank. The mean mark for Section A was 63%, and the standard deviation was 14%.

Section B asked you to provide a detailed sketch of a thin section showing its mineralogy and texture. To do well on this question, answers needed to provide observations justifying the mineral identification, a sketch showing how the crystals fit together in the rock texture, and a short narrative describing the overall texture. The best answers showed specific mineral textures on their sketches, which they could then refer to when answering later parts of the question. The most common mistake was not providing a tectonic setting. The mean mark for Section B was 58%, and the standard deviation was 20%. The low mean mark is because some students ran out of time and did not answer parts of the question.

Section C asked you to use written information to classify some igneous rocks, and asked you to demonstrate a working knowledge of mineral crystallography, symmetry, and the silicate mineral groups. The classification part of the question was overall well handled; the most common mistake was forgetting to write a sentence justifying your choice of classification scheme. The mineralogy part of the question was less confidently handled and a number of students did not attempt to answer it. The mean mark for Section C was 55%, and the standard deviation was 19%.

The graphs at the top of the next page show the marks distribution for each question.



Some general feed-forward for future assessments in this and other units

- If you are given a question where there is an obvious choice of answers (e.g., to state whether the shortest lattice dimension is *a, b* or *c*), it is always worthwhile writing something down – you may end up being right and picking up some extra marks.
- If you are asked to do something specific in a question (e.g., to write one sentence justifying your choice of classification scheme), there will be marks available for doing this – so it is worth writing something down, even if you are unsure.
- Show your working or write down your thought process, especially if you are not sure that you know "the answer" to the question – you may still pick up marks for an incomplete answer.
- Time management is important in assessments. On a 2-hour test with 100 marks available, the time budget is about 1 mark per minute: 10 minutes at the start for reading the question paper; then 100 minutes to answer the questions; then 10 minutes at the end to check your work and add any final details. It is worthwhile practicing skills under timed conditions so that you can get a feel for how much you are able to draw or write down in 10 minutes, or 15 minutes, or 20 minutes. The people marking your work know that you are working under time pressure, and we take this into account. For this reason it is usually better to attempt to answer every question on the paper, rather than devoting all of your attention to just one question and leaving the rest unanswered.

Some specific feed-forward for the EART22201 exam

- Take care when describing melting and crystallization. The minerals in rocks that have been erupted or intruded **do not** crystallize from peridotite. Peridotite (mantle) melts to create mafic magmas. Those mafic magmas then undergo fractional crystallization to create the diversity of rock types that we see in the Earth's crust.
- Take care with intrusive vs extrusive rock names. Example: rhyolite and granite have the same chemical composition, but rhyolites are extrusive (fine-grained) and granites are intrusive (coarse-grained). So if you see a fine-grained silica-rich volcanic rock, it cannot be a granite!
- Check that you know and can explain the difference between S-type and I-type granites.
- Check that you know and can describe some common igneous textures, and can sketch their overall appearance. (Remember that in the January exam you will not have the Practical Reference Handout to rely on.)
- If you are going to draw a diagram, please make it a large diagram, and make sure that your writing is large enough and neat enough for me to read easily.

I hope you have a good rest over the Christmas break, and all the best in the January exams.

Margaret